

Alekhya Dabbiru

Seattle, WA | alekhyadabbiru19@gmail.com | (206) 250-1686 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)
DATA ANALYST | SQL | PYTHON | TABLEAU | POWER BI | STAKEHOLDER MANAGEMENT

SUMMARY

Data Science graduate with 2+ years of experience leveraging Python, SQL, ETL, and business intelligence to automate reporting, improve data quality, and guide product and operational decisions. Skilled in KPI tracking, exploratory data analysis, A/B testing, Snowflake, Tableau, and Power BI, delivering reproducible analyses and executive-ready insights.

SKILLS

- **Programming and Libraries:** Python, SQL, R, Pandas, NumPy, SciPy, Scikit-learn, XGBoost, CatBoost
- **Machine Learning and Statistics:** Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Random Forest, NLP, TF-IDF, K-Means, Deep Learning, Time-Series Forecasting, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation, Hypothesis Testing, A/B Testing, Threshold Optimization
- **Data Engineering and Cloud:** ETL and ELT Pipelines, Data Warehousing, Data Modeling, Star Schema, Dimensional Modeling, Data Validation, Data Quality Monitoring, Data Transformation, Snowflake, Databricks, Spark, AWS S3, Microsoft Azure
- **Analytics and Business Intelligence:** EDA, Descriptive Statistics, Predictive Analytics, Ad Hoc Analysis, Trend Analysis, KPI Tracking, Churn Analysis, Retention Metrics, Forecasting, Tableau, Power BI, Excel, Dashboard Development, Data Storytelling
- **Tools and Collaboration:** Docker, Git, GitHub, Jupyter Notebook, Version Control, Reproducible Pipelines, Stakeholder Management, Cross-Functional Collaboration, Requirements Gathering, Executive Reporting

EXPERIENCE

Graduate Data Scientist | Tacoma Power

Jan 2026 - Jun 2026

- Built Python, SQL, and Snowflake ETL pipelines for **150K+ inspection records**, standardizing multi-year contractor data, cutting processing time **30%**, and raising inspection-data reliability for predictive maintenance.
- Applied predictive analytics and anomaly detection to rank high-risk assets, guide risk-based maintenance, and strengthen outage-prevention planning.
- Automated Tableau dashboards, KPI tracking, and data-quality monitoring, reducing manual reporting **40%** and giving cross-functional stakeholders clearer visibility into pipeline health, asset risk, and maintenance priorities.

Data Analyst | Manuh Global Technologies Pvt Ltd

Nov 2023 - Aug 2024

- Built Python and SQL validation, preprocessing, and transformation workflows across **60K+ records**, increasing downstream data quality **25%** for predictive modeling, rapid analysis, and recurring reporting.
- Automated extraction and optimized SQL queries across **7+ datasets**, cutting report turnaround **20%** and accelerating forecasting, trend analysis, root-cause reviews, and business planning.
- Developed Power BI and Tableau dashboards for KPI tracking, executive reporting, and data storytelling, increasing stakeholder adoption **30%** and clarifying model inputs, performance trends, and business outcomes.

Data Analyst | Edumoon, Andhra University

Nov 2022 - Oct 2023

- Analyzed engagement and retention across **7 campuses** using segmentation, cohort analysis, and KPI reporting, contributing to a **10-12% increase in active platform adoption** and informing campus outreach.
- Built a centralized data pipeline for multi-campus integration, reducing manual processing and retrieval time **35-40%** and accelerating retention, product, and feature analysis across campus programs.
- Created stakeholder-ready dashboards and executive reports to surface adoption, engagement, and feature-performance trends, strengthening outreach prioritization and cross-functional planning for campus growth.

PROJECTS

[Parking Flow Intelligence](#) | Python, Pandas, Scikit-learn, Random Forest, Time-Series Forecasting

- Developed forecasting models with lag features, rolling averages, and Random Forest across **35K+ records from 30 locations**, improving R-squared from **0.874 to 0.974** with **1.71% MAE** and **3.14% RMSE**.
- Engineered a reusable ETL workflow with data-quality checks, feature engineering, model evaluation, and automated reporting, identifying **5,701 high-risk occupancy observations** and guiding congestion monitoring across 30 locations.

[FetchFood Penalty Optimization](#) | Python, A/B Testing, SciPy, Scikit-learn, XGBoost, CatBoost, Joblib

- Designed an A/B experiment across three fee groups using z-tests and profit analysis, identifying a **\$20 penalty** as the strongest strategy with a **10.03% cancellation rate** and **\$14.05 average profit per order**.
- Tuned Random Forest, XGBoost, and CatBoost to **0.78 precision** and serialized reproducible pipelines with Joblib, enabling repeatable inference, version-controlled artifacts, and consistent model evaluation.

[Airline Delay Control Tower ETL](#) | Python, SQL, SQLite, ETL, Star Schema, Data Quality Monitoring

- Built a Python, SQL, and SQLite ETL pipeline for **500K airline records** with automated data-quality checks and star schema modeling, creating **5 SQL-ready fact and dimension tables** for repeatable operational analytics.
- Ranked **2,777 routes** by delay frequency, cancellation risk, and traffic volume to generate recovery-priority reports for capacity planning, performance monitoring, and airline operations.

[AI Incident Risk Analyzer](#) | Python, NLP, TF-IDF, K-Means, Truncated SVD, Text Analytics

- Applied NLP preprocessing and TF-IDF vectorization to categorize **1,465 AI incidents** across safety, privacy, governance, bias, and misinformation risk domains.
- Flagged **145 high-severity incidents** and discovered **8 risk clusters** using severity scoring, K-Means, Truncated SVD, and cosine-similarity search.

CERTIFICATIONS

- **Microsoft Azure AI Fundamentals** - Microsoft
- **Generative AI Fundamentals** - Databricks
- **Artificial Intelligence A-Z 2025 and Machine Learning A-Z** - Udemy

EDUCATION

Master of Science in Data Science | Seattle University, Seattle, WA

Sept 2024 - Jun 2026

Relevant Coursework: Statistical Modeling, Machine Learning, Database Management Systems, Big Data, Deep Learning

Bachelor of Engineering in Computer Science | Andhra University, Visakhapatnam, India

Jul 2020 - May 2024